

JEE MAIN REVISION MASTER: FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Types of Functions: One-One, Onto & Bijective	22 PYQs	Function Classification
High	Composition of Functions & Invertibility	14 PYQs	Function Operations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Types of Functions: One-One, Onto & Bijective	$\text{Total Mappings } A \rightarrow B = n^m$, $\text{Injections } (m \leq n) = {}^n P_m$, $\text{Bijections } (m=n) = n!$
Composition of Functions & Invertibility	$(f \circ g)^{-1} = g^{-1} \circ f^{-1}$, $f(x) = \frac{ax+b}{cx-a} \implies f^{-1}(x) = \frac{bx+a}{cx-a}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Types of Functions: One-One, Onto & Bijective (Very High Yield — 22 PYQs)

Core Principle: Function mappings from set A (m elements) to set B (n elements): Total functions = n^m . Injective (one-one): $f(x_1) = f(x_2) \implies x_1 = x_2$ (possible only if $m \leq n$; total count = ${}^n P_m$). Surjective (onto): Range = Codomain (possible only if $m \geq n$). Bijective (one-one and onto): $m = n$, total bijections = $n!$.

■ **Caution / JEE Pitfall:** A strictly monotonic function ($f'(x) > 0$ or $f'(x) < 0$ throughout its domain) is always one-one (injective).

■ **Speed Trick:** Horizontal Line Test: If any horizontal line intersects the graph at most once, the function is one-one.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $f(x)=(x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ is..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "The domain of the function $\sqrt{(x-7)(9-x)}$ is..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "Let the functions 'f' and 'g' be $f:[0,(\pi/2)] \rightarrow R$ given by $f(x)=\sin x$ and $g:[0,(\pi/2)] \rightarrow R$ given by $g(x)=\cos x$. Then $f \circ g$ is..." [Ans: N/A]

3.2 Composition of Functions & Invertibility (High Yield — 14 PYQs)

Core Principle: Composition $(g \circ f)(x) = g(f(x))$. A function $f: A \rightarrow B$ is invertible if and only if f is BIJECTIVE (one-one and onto). If $f(x) = y$, then $f^{-1}(y) = x$. Binary operations: Commutative if $a * b = b * a$; Associative if $(a * b) * c = a * (b * c)$; Identity element e satisfies $a * e = e * a = a$.

■ **Caution / JEE Pitfall:** Composition is NOT commutative in general: $f \circ g \neq g \circ f$!

■ **Speed Trick:** If $f(x) = (ax+b)/(cx-a)$ where the diagonal coefficients are equal and opposite (a and $-a$), then $f(f(x)) = x$ and $f^{-1}(x) = f(x)$!

Representative Source PYQs:

- [2025 Shift] Q#4 (MCQ): "If $f(x) = \sin[\pi^2]x - \sin[-\pi^2]x$, where $[x]$ = greatest integer $\leq x$, then which of the following is not true?..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "Let $(g \circ f)(x) = \sin x$ and $f \circ g(x) = (\sin \sqrt{x})^2$. Then,..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "Let $f(x) = \sin 2x + \cos 2x$ and $g(x) = x^2 - 1$ then $g(f(x))$ is invertible in the domain..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$f(x) = (x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ is

- (A) $-\sqrt{x}+1$
- (B) $\sqrt{x}-1$
- (C) $\sqrt{x}+1$ [Correct]
- (D) $\sqrt{x-1}$

Correct Answer: C

Solution Analysis: The graph of $g(x)$ is the reflection of the graph of $f(x)$ in the line $y=x$. Reflection in the line $y=x$ gives the inverse relation of the function. Given, $f(x) = (x-1)^2, x \geq 1$. Let $y = (x-1)^2$. Since $x \geq 1$, we have $x-1 \geq 0$. So when we take square root, $\sqrt{y} = x-1$. Hence, $x = \sqrt{y} + 1$. Now interchange x and y to get the reflected function: $y = \sqrt{x} + 1$. So, $g(x) = \sqrt{x} + 1$. Hence the correct option is Option C $\sqrt{x} + 1$.

Benchmark Problem 2: KCET 2024 (Q#4)

Let $(g \circ f)(x) = \sin x$ and $f \circ g(x) = (\sin \sqrt{x})^2$. Then,

- (A) $f(x) = \sin^2 x, g(x) = x$
- (B) $f(x) = \sin \sqrt{x}, g(x) = \sqrt{x}$
- (C) $f(x) = \sin^2 x, g(x) = \sqrt{x}$
- (D) $f(x) = \sin \sqrt{x}, g(x) = x^2$

Correct Answer: None

Solution Analysis: ■ $g(f(x)) = \sin x$ and $f(g(x)) = (\sin \sqrt{x})^2$ (a) $f(x) = \sin^2 x$ and $g(x) = x$. Now, $f(g(x)) = f(x) = \sin^2 x$ and $g(f(x)) = g(\sin^2 x) = \sin^2 x$. (b) $f(x) = \sin \sqrt{x}$ and $g(x) = \sqrt{x}$. Now, $f(g(x)) = f(\sqrt{x}) = \sin \sqrt{\sqrt{x}} = \sin(x^{1/4})$ and $g(f(x)) = g(\sin \sqrt{x}) = \sqrt{\sin \sqrt{x}}$. (c) $f(x) = \sin^2 x$ and $g(x) = \sqrt{x}$. Now, $f(g(x)) = f(\sqrt{x}) = \sin^2 \sqrt{x}$ and $g(f(x)) = g(\sin^2 x) = \sqrt{\sin^2 x} = |\sin x|$. (d) $f(x) = \sin \sqrt{x}$ and $g(x) = x^2$. Now, $f(g(x)) = f(x^2) = \sin \sqrt{x^2} = \sin |x|$ and $g(f(x)) = g(\sin \sqrt{x}) = (\sin \sqrt{x})^2 = \sin^2 \sqrt{x}$.